



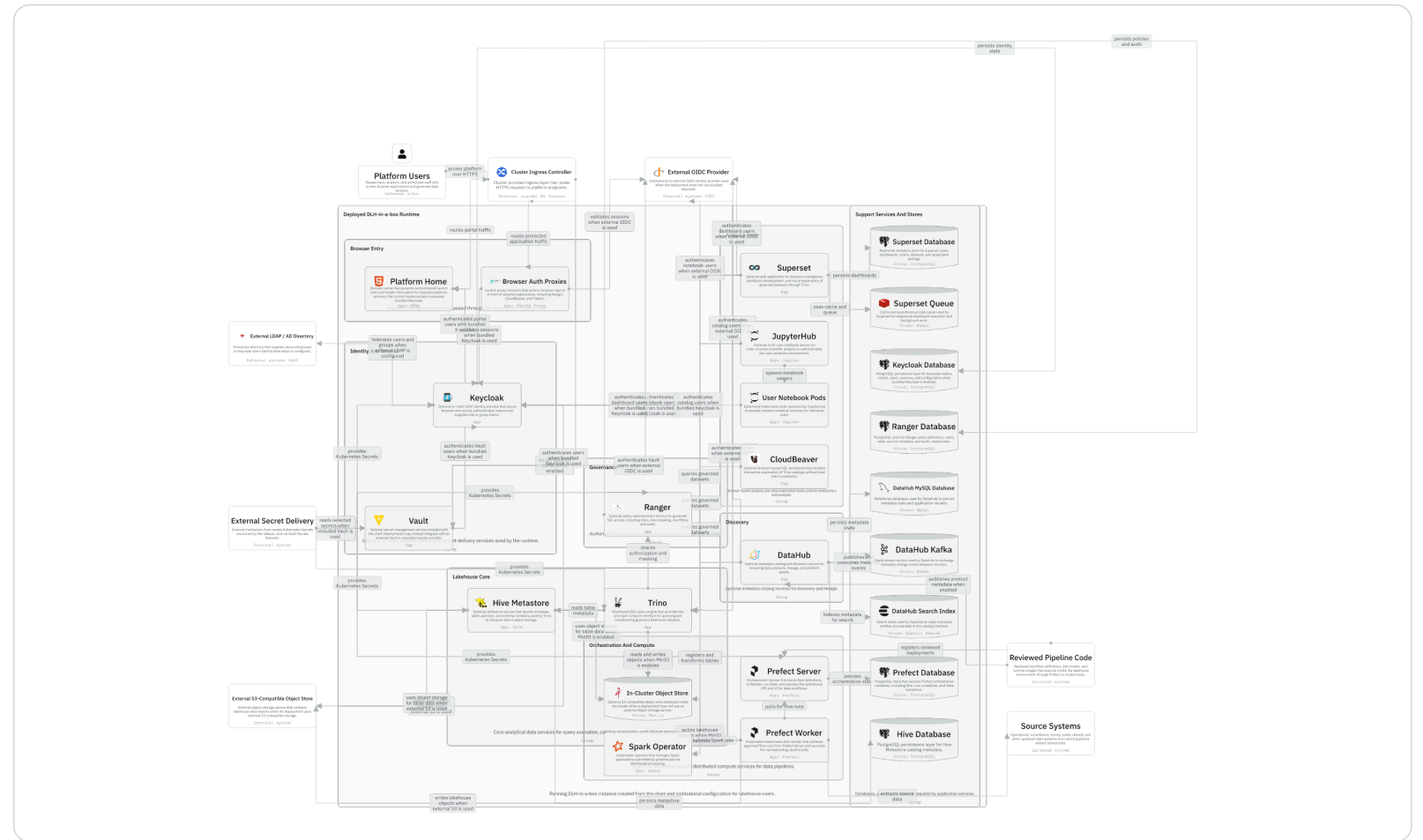
## Session 02 / C4 and IcePanel handover

# Architecture Model And Runtime Topology

One hour handover deck

40-45 min narrative / 10-15 min checkpoint /  
5 min Q&A

The model separates chart product, published package, consumer configuration, and deployed runtime.



# Session path

---

- 1 **Modelling rules**
- 2 **Packaging view**
- 3 **Runtime view**
- 4 **How diagrams support changes**

## Session rhythm

40-45 min narrative  
10-15 min checkpoint  
5 min Q&A

**Presenter notes carry the script and demo prompts.**

# The model is scoped to the reusable chart

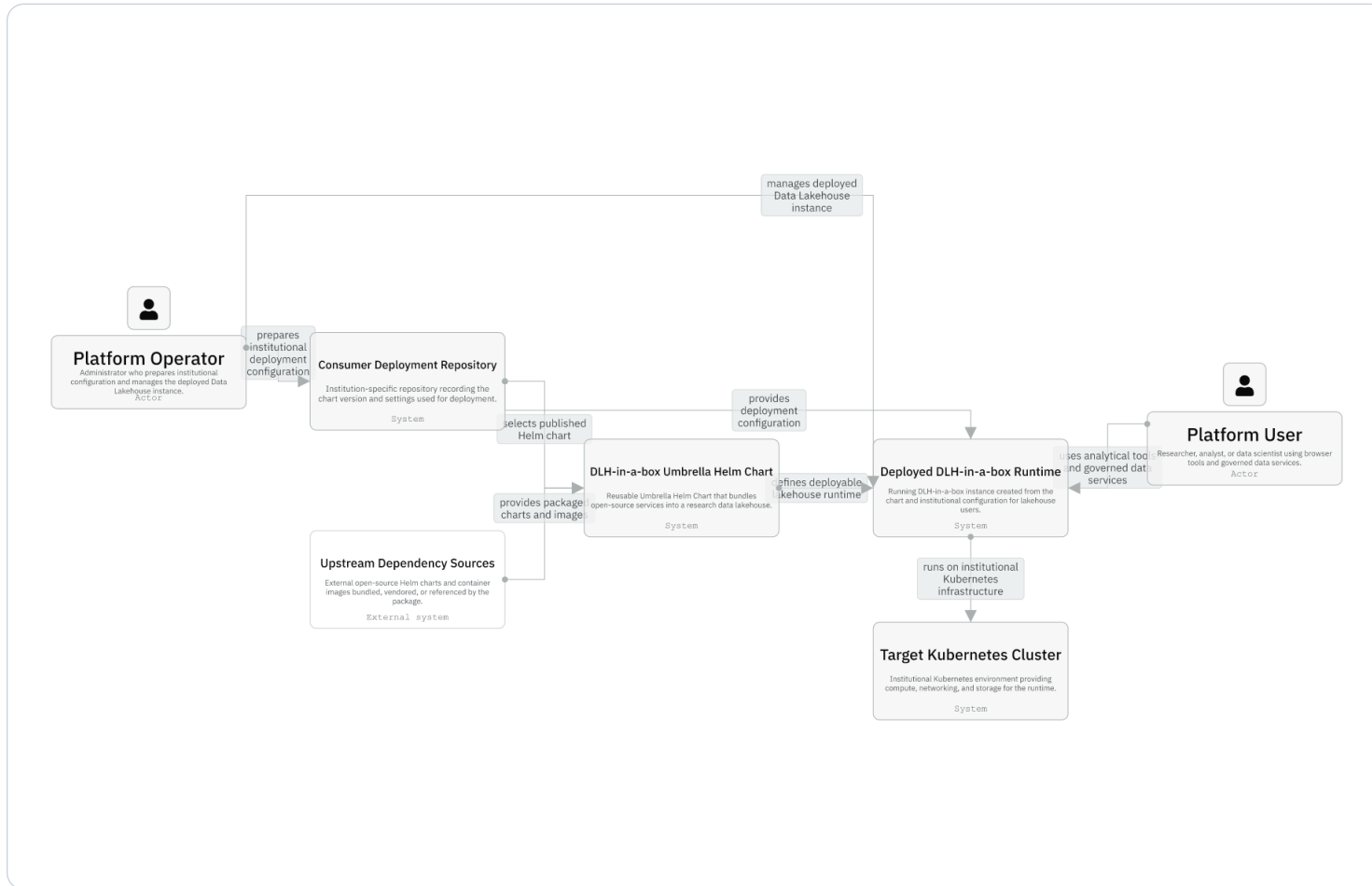
---

## What belongs in this architecture view

- DLH-in-a-box as a Helm chart product.
- The generic runtime instantiated by the chart.
- Upstream dependency sources and packaged artifacts.
- Owned component diagrams for chart source, Hive, Trino patch points, validation, and publish automation.

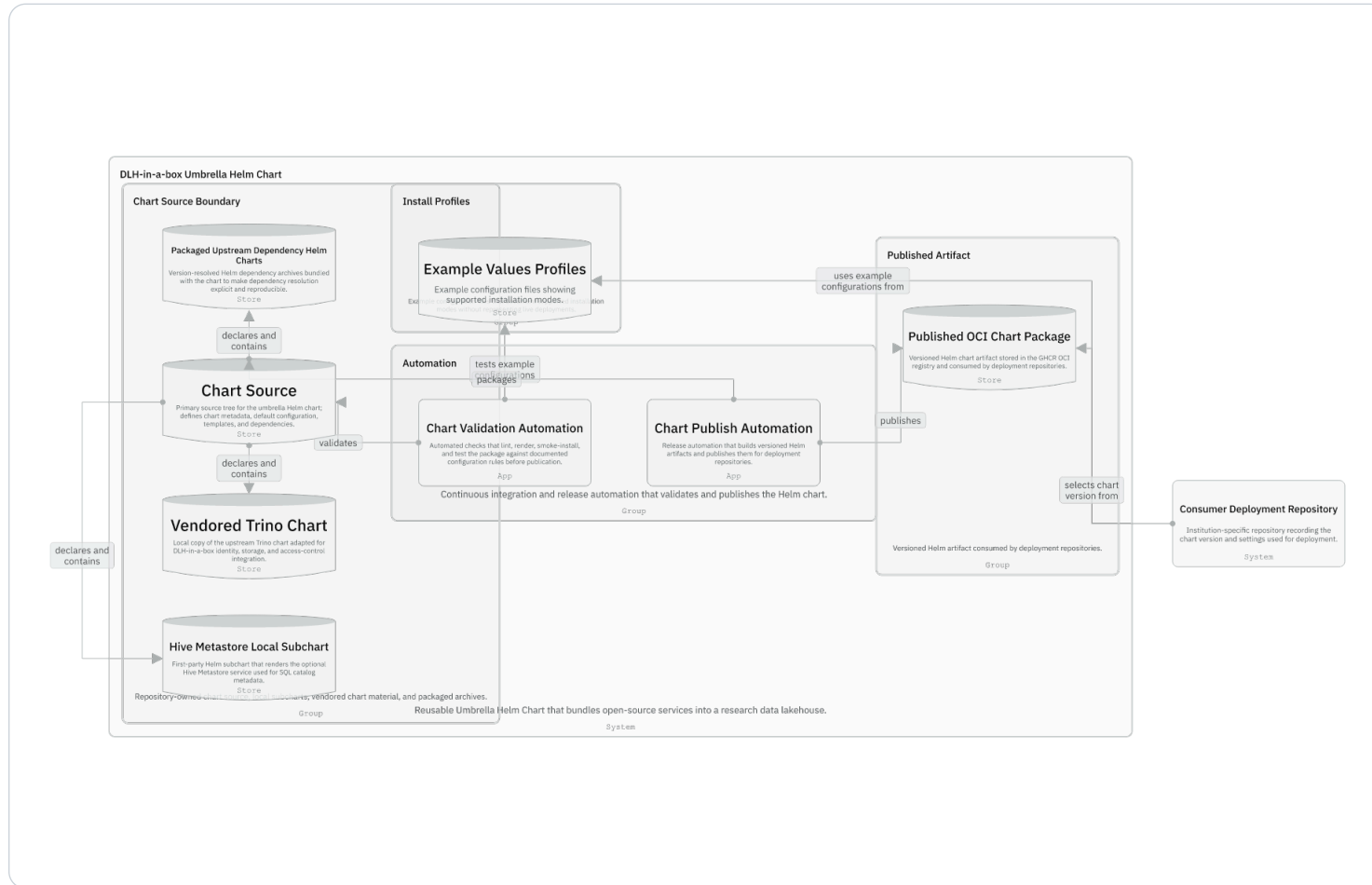
**DLH-in-a-box  
as a Helm  
chart  
product.**

# Level 1: context and consumers



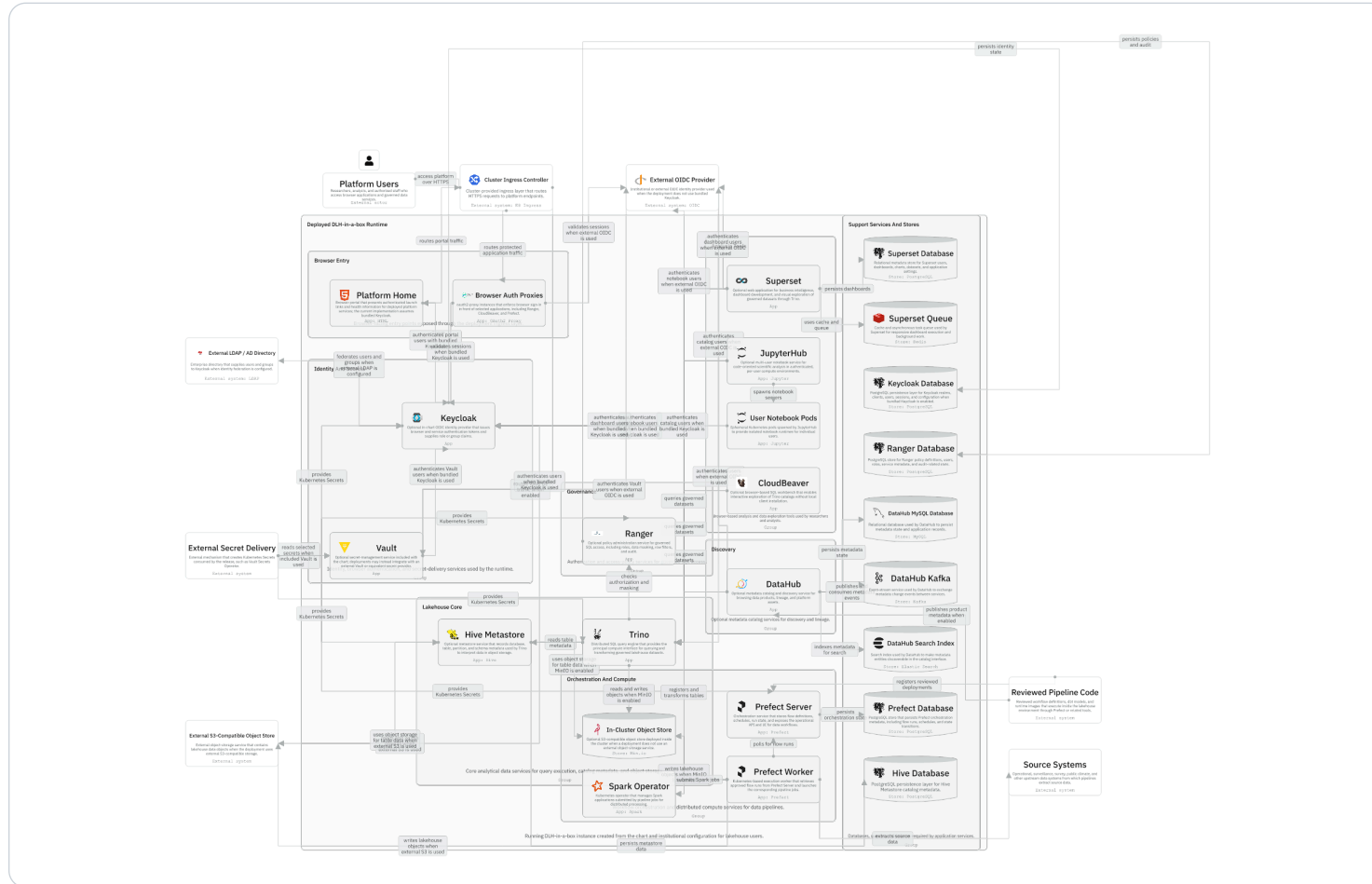
- Consumer repositories select chart versions.
- Institutional settings live outside the chart product.
- The runtime is the system operators and users experience.

# Level 2A: chart product and packaging



- Chart source declares dependencies and templates.
- Packaged upstream archives make resolution reproducible.
- Validation and publish automation protect the package before release.
- The OCI package is what deployment repos consume.

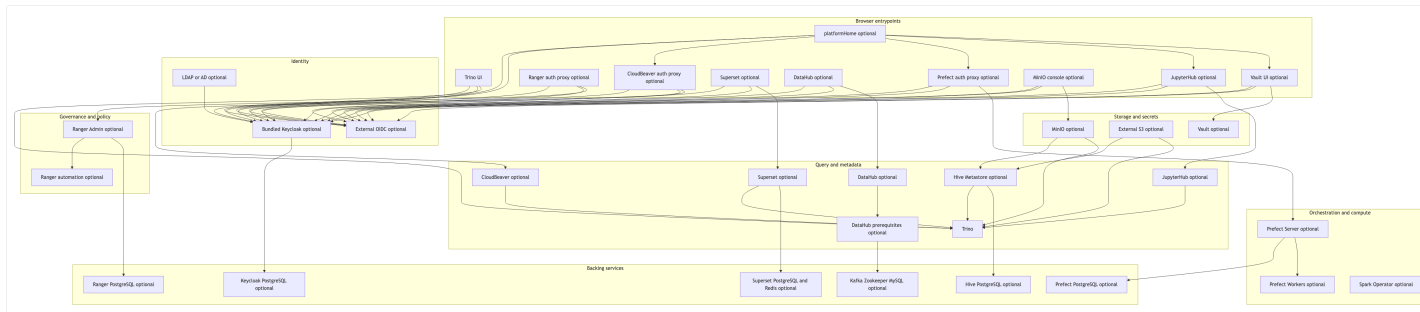
# Level 2B: runtime deployed by the chart



- Trino sits at the centre of the query plane.
- Identity and browser entypoints sit above runtime tools.
- Prefect, Spark, and optional apps surround the core data path.
- Support stores belong near the services that own them.

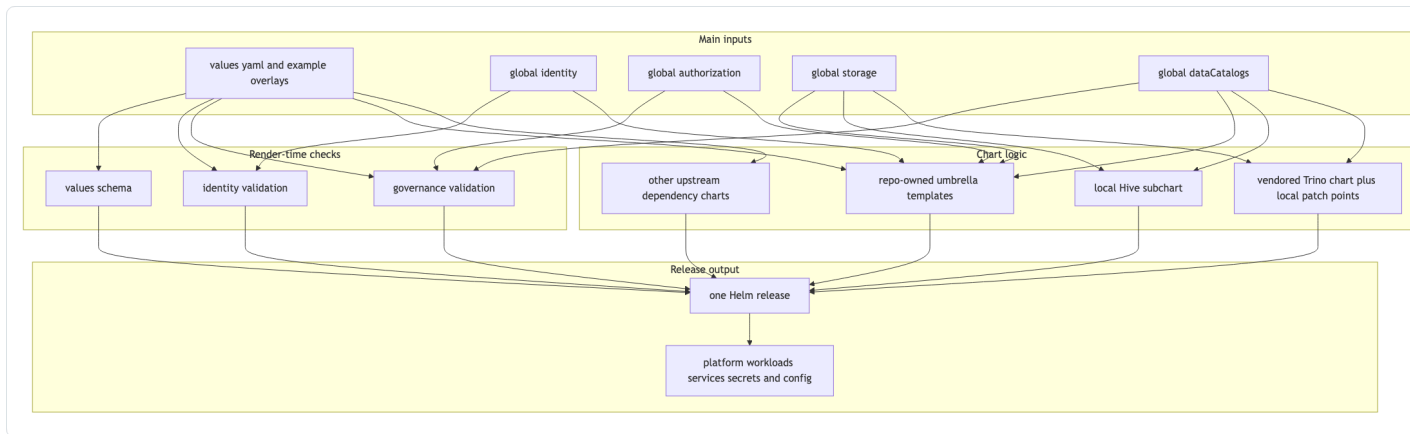
# Manual view: everything enabled

- The chart is modular; not every install enables every component.
- Browser entrypoints, identity, governance, query, orchestration, and storage are separate lanes.
- Optional services are still part of the model because examples can turn them on.



# How the chart assembles the runtime

- Values drive validation, templates, dependency charts, and rendered resources.
- Example overlays exercise supported combinations.
- CI tests rendering and packaging rather than pretending one deployment shape covers all cases.





# Use the right diagram for the question

---

| Question                          | Best diagram                      |
|-----------------------------------|-----------------------------------|
| Who consumes the chart?           | Context                           |
| What gets packaged and published? | Chart product and packaging       |
| What runs in a cluster?           | Runtime deployed by the chart     |
| Where is local chart logic?       | Chart source components           |
| How is CI or publish wired?       | Validation and publish automation |

# Guided checkpoint: trace one feature through the model

---

```
Open docs/architecture/dlh-in-a-box-icepanel-model.md
```

```
Find the Level 2 runtime object for Trino
```

```
Follow its relationships to Hive, storage, identity, and Ranger
```

## Checkpoint questions

- Can the developer explain whether the change belongs in chart source, a subchart, or consumer settings?
- Can they name the external systems versus repo-owned material?

# Session 02 landing

---

## What the inheriting team should now be able to do

- They know why there are separate product and runtime views.
- They can choose the diagram that answers a maintainer question.
- They understand why deployment-specific infrastructure is mostly outside this model.

### Leave-behind

Use the PPTX notes as a presenter script and source-map.md as the reference trail.